Md Kamrul Islam

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Google Scholar | ORCID | Portfolio | GitHub | LinkedIn

Research Interests

Foundation Models, Agentic AI, Retrieval-Augmented Generation, Information Retrieval, and Efficient Language Models

Education

Erasmus Mundus Joint Master's Degree in Big Data Management and Analytics (BDMA) *Sep 2023 – Oct 2025*
CentraleSupélec | Universitat Politècnica de Catalunya | Université libre de Bruxelles *France | Spain | Belgium*

- **Grade:** 14.5/20 (Mention Bien).
- **Thesis:** *A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models* (supervisor: Dr. Tiphaine Henry, CEA-LIST), which led to the first-author BPM 2026 paper.
- **Scholarship:** Erasmus Mundus Scholarship, European Commission (Fully funded).

Bachelor of Engineering in Software Engineering *Mar 2018 – Dec 2021*
Sichuan University *Chengdu, China*

- **CGPA:** 3.64/4.00 (87%, Top 5%).
- **Thesis:** *Brain Tumor Detection and Classification Using Convolutional Neural Networks*. [Code]

Research Experience

Research Collaborator (Remote) *Oct 2026 – Present*
Collaborating with Prof. Mattia Salnitri, University of Bergamo *Bergamo, Italy*

- Developing an Agentic framework that uses goal-oriented requirement engineering to drive generation and refinement of security-annotated business process models, with human-in-the-loop feedback and downstream code generation and execution. ***(Manuscript in preparation)***

AI Research Engineer Intern [Preprint] [DOI] *May 2025 – Oct 2025*
LISSI, Université Paris-Est Créteil | in collaboration with CEA–LIST *Paris, France*

- Developed a hybrid LLM-based framework integrating semantic extraction, RAG, and rule-based validation to automate security annotation generation in business process models.
- Implemented BM25–FAISS retrieval and curated 27 SecBPMN2 text pairs with expert annotations for evaluation.
- Achieved 0.58 precision and 0.52 recall, compared with human baselines of 0.29 and 0.50, respectively, while reducing annotation time by approximately 95%.

Tech Stack: *Python, LLMs, RAG, BM25, FAISS, LangChain*

Graduate Research Assistant [Code] *Oct 2024 – Mar 2025*
Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), CentraleSupélec *Paris, France*

- Developed an unsupervised framework based on Group-Equivariant CNNs (G-CNNs) for chest X-ray clustering.
- Explored geometric equivariance to improve feature robustness and reduce reliance on data augmentation.
- Implemented GPU-accelerated training for large-scale medical imaging experiments.

Tech Stack: *PyTorch, OpenCV, CUDA, HPC*

Teaching Experience

Lecturer [Faculty Profile] *Apr 2026 – Present*
Department of Computer Science, American International University-Bangladesh (AIUB) *Dhaka, Bangladesh*

- Teach undergraduate courses in Python Programming, Data Processing, Data Visualization, and Compiler Design; develop course plans and materials following Outcome-Based Education (OBE) principles.
- Contribute to curriculum development and academic planning for the BSc in Data Science program.
- Mentor undergraduate thesis students on research formulation, methodology design, and implementation.

Publications

1. **M. K. Islam**, T. Henry, M. Salnitri, J. Köpke, and S. Souihi, "A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models," in *Proceedings of the 24th International Conference on Business Process Management (BPM 2026)*, Toronto, ON, Canada, 2026, pp. 168–184. *Lecture Notes in Business Information Processing*, vol. 599, Springer, Cham. [DOI]
2. C. K. Sah, L. Xiaoli, M. M. Islam, and **M. K. Islam**, "Navigating the AI Frontier: A Critical Literature Review on Integrating Artificial Intelligence into Software Engineering Education," in *Proceedings of the 36th International Conference on Software Engineering Education and Training (CSEE&T 2024)*, Würzburg, Germany, 2024, pp. 1–5. [DOI]

Presentations and Posters

M. K. Islam, X. Zhuang, J. Ma, and A. Malhotra, "Unsupervised Chest X-ray Clustering with Group Equivariant Convolutions," (poster presentation) *RAMH 2026: Recent Advances in Machine Learning for Healthcare*, Paris, France, 2026. [HAL Record] [Poster]

Professional Experience

Software Engineer Intern *Dec 2020 – May 2021*
Chengdu Suncape Co., Ltd. *Chengdu, China*

- Built distributed data pipelines for large-scale data processing, from ingestion to analytics, and worked with an Agile Scrum team on modular development and CI/CD practices.

Tech Stack: *Apache Spark, Scikit-learn, SQL, CI/CD*

Technical Skills

Programming Languages: Python, C/C++, SQL

ML Frameworks & Libraries: Scikit-learn, PyTorch, OpenCV

LLMs & Agentic AI: RAG, LangChain, LangGraph

Data Engineering & Knowledge Graphs: Apache Spark, PostgreSQL, Neo4j, SPARQL

Tools & Computing: Git, CUDA, HPC

Selected Projects

Purchase Intelligence [Code] *Mar 2026*
Developed an agent AI system that analyzes Amazon customer reviews to generate interpretable recommendations; built for the Major League Hacking (MLH) challenge.

Tech Stack: *JavaScript, Node.js, LLMs APIs, AI Agents, Express.js*

DigiScan360 [Code] *Feb 2024 – Jun 2024*
Developed an LLM-powered competitive intelligence prototype using big data and knowledge graphs.

Tech Stack: *PySpark, LLMs, Microsoft Fabric, Power BI, GraphDB, SPARQL*

Academic Service & Activities

BanglaKnight Shared Task 1, BLP Workshop at NAACL 2027 [Leaderboard] *Nov 2026 – Present*
Team Bornomala | *Working on adversarial prompt classification across Bangla, romanized Bangla, and code-mixed text* ***(Manuscript in preparation)***

Reviewer, 4th International Conference on Computing Advancements, 2026 [Certificate] *Aug 2026*
ACM SIGAPP In-Cooperation | American International University-Bangladesh *Dhaka, Bangladesh*

Twelfth European Big Data Management & Analytics Summer School (eBISS 2024) [Poster] *Jul 2024*
University of Padova *Padova, Italy*

Languages

English: Proficient | **Chinese:** Upper-intermediate | **French:** Elementary | **Bengali:** Native

References

References available upon request.